

Answer any three questions (including subparts)

1 a) Fill in the blanks (with the correct answer) for the following:

i) Let an array be represented as $A(2:20, -3:30, -1:25, 0:25, 5:30)$. Assuming row major representation of the array (using sequential byte addressing scheme) with 2 bytes per element. If m is the address of $A(4, -1, 6, 7, 12)$ then the address of the element represented as $A(3, 10, 15, 6, 15)$ is _____

ii) If a polynomial is represented in the form $(m, e_{m-1}, b_{m-1}, e_{m-2}, b_{m-2}, \dots, e_0, b_0)$ where, the first entry m represents the number of non-zero terms followed by two entries (e_i, b_i) for each non-zero term (representing an exponent-coefficient pair) in the order $e_{m-1} > e_{m-2} > \dots > e_0 > = 0$. The total number of element locations needed to store the result using the above representation when the polynomial $x^{1000} - 3x^{200} + 3x^5 - 2x^4 + 5x + 2$ is added to the polynomial $x^{500} + 3x^{200} + 9x^4 + 2x^3 + 10x + 1$ is _____

b) State whether the following statements are true or false:

i) The symbolic language used by the traffic police person to control the traffic can be considered as software

ii) In Data Structure arrays are always implemented by using consecutive memory locations.

(6+1+3)

2 a) What is the difference between direct and indirect recursion? What is tail recursion? Explain with suitable example how advantageous tail recursion could be in saving memory space as compared to the recursion which is not tail recursive.

b) Explain with suitable examples (specifying the names of programming languages which follow 0 - based indexing, 1- based indexing, n - based indexing) 0 - based indexing, 1- based indexing and n-based indexing for arrays.

c) Write your algorithms (using C pseudocode) for the menu driven implementation to store a list of 15 elements (taken as input from the user) in an array and perform any of the following task depending upon the user's choice:

i) Take an element from the user as input and insert it at the n^{th} location of the list (where the value of n is taken as input from the user).

ii) Take an element from the user as input and delete the element from the list if the element is found in the list otherwise it displays "Element not found".

(2.5+4.5+3)

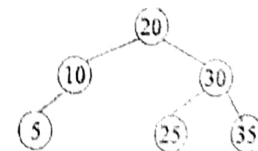
- 3 a) Let a sparse matrix be stored in an array $X(0:t, 0:2)$ where t is the number of nonzero elements. The elements $X(0,0)$ and $X(0,1)$ contain the total the number of rows and the total number of columns of the matrix respectively. $X(0,2)$ contains the total number of nonzero elements of the matrix. Let $X(i,0)$, $X(i,1)$ and $X(i,2)$ contain row number, column number and the value of the nonzero element in the matrix respectively, for $1 \leq i \leq t$. Write algorithm(s) (using C pseudocode) to store two sparse matrices (taken as input from the user) in the form (as specified above using X), perform addition of these matrices, store the result in the form (as specified above using X) and display the result (in the form which is usually understood by an user unfamiliar with the representation specified using X).
- b) Write your algorithm(s) (using C pseudocode) to provide solution (with minimal number of disk moves) to the "Tower of Hanoi" problem with n number of disks (where the value of n is provided by the user as input). (6+4)
- 4 a) Write algorithm(s) (using C pseudocode) to perform the subtraction of two polynomials [where each polynomial is stored the array as a list of elements in the form $(m, e_{m-1}, b_{m-1}, e_{m-2}, b_{m-2}, \dots, e_0, b_0)$ where, the first entry m represents the number of non-zero terms followed by two entries (e_i, b_i) for each non-zero term (representing an exponent-coefficient pair) in the order $e_{m-1} > e_{m-2} > \dots > e_0 > = 0$] and print the resultant polynomial.
- b) Write down algorithm(s) (using C pseudocode) to perform the following operations on a stack of size n (implemented using array) depending on users choice (where the value of n is provided by the user as input)
- Push k number of elements provided by the user as input to the stack (where the value of k is provided by the user as input)
 - Pop an element from the stack and print the popped (deleted) element
 - Display the contents of the stack
- (4+6)
- 5 a) Write non-recursive, tail recursive and recursive (but not non-tail recursive) algorithms (using C pseudocode) to find the factorial of a given number (provided by the user as input).
- b) A complex-valued matrix X is represented by a pair of matrices (A,B) where A and B contain real values. Write algorithm(s) (using C pseudocode) which computes the product of two complex valued matrices (A,B) and (C,D) , where $(A,B) * (C,D) = (A + iB) * (C + iD) = (AC - BD) + i(AD + BC)$. Determine the number of additions and multiplications if the matrices are all $n \times n$ (4.5+5.5)
- 6 a) Write algorithm(s) (using C pseudocode) to reverse the elements of a stack using stack and no other data structures.
- b) Write algorithm(s) (using C pseudocode) to swap the values of 2 integer variables without using another/third/temporary variable.

(7+3)

Time: 3 hrs.

Attempt any five questions (including subparts)

- 1 a) Obtain an addressing formula for the element $A(i_1, i_2, \dots, i_n)$ in an array declared as $A(l_1:u_1, l_2:u_2, \dots, l_n:u_n)$. Assume a sequential column major representation of the array with m bytes per element and a as the address of $A(l_1+1, l_2+2, l_3+3, \dots, l_n+n)$ where $l_k+k \leq u_k, 1 \leq k \leq n$.
- b) Write an algorithm (using C pseudocode) to implement a priority queue where the element having the minimum value is given the maximum priority.
- c) Write recursive algorithms (using C pseudocode) for inorder and postorder traversal of a binary tree (where the tree is implemented using array) and specify their space and time complexities.
- (3+4+3)
- 2 a) Write algorithm(s) (using C pseudocode) to insert an element into a circular-queue where the circular-queue is implemented using a doubly linked list. Specify its time complexity using Θ notation.
- b) State whether the following are true or false.
- i) $7 \cdot 2^n + n^2 = O(2^n/6)$ ii) $5 \cdot 2^n + n^2 = \Omega(n^{10000})$
- iii) $6n^2 + 4n + 2 = \Theta(2^n)$ iv) $7n^2 + 4n + 2 = O(n^{2.5})$
- v) $n/250000 = \Omega(500000)$ vi) $500005 = \Theta(5)$
- c) Write an algorithm (using C pseudocode) to reverse the direction of the links of a singly linked list during traversal.
- (2+3+5)
- 3 a) Specify the postorder sequence for the binary tree whose inorder and preorder sequences are as follows:-
Inorder: **B C A E D G H F I** Preorder: **A B C D E F G H I**
- b) Write an algorithm (using C pseudocode) to print the data present in all the nodes of a binary tree (having n nodes), level by level. Within a level the data (present in the nodes) are to be printed from the left node to the right node. What is the time complexity of your algorithm?
- c) Prove that for any nonempty binary tree, T , if n_0 is the number of terminal nodes and n_2 is the number of nodes of degree 2, then prove that $n_0 = n_2 + 1$.
- (3+4+3)
- 4 a) If $f(n) = a_m n^m + \dots + a_1 n + a_0$ where $a_0, a_1, \dots, a_m$ are constants then prove that $f(n) = \Theta(n^m)$, for $n \geq 1$ and $a_m > 0$.
- b) Write an algorithm (using C pseudocode) to search a key in a Binary Search Tree (BST) and delete it such that the resulting tree is a BST. Can the same algorithm be used for deleting an element in an AVL tree such that the resulting tree is an AVL tree? Specify the worst case time complexity to search a key in a BST and in an AVL tree using Θ notation.
- c) Explain the advantages and disadvantages of using AVL trees. Explain why the time complexity analysis of our algorithm using asymptotic notations may fail when we use disk storage.
- (3+5+2)
- 5 a) Write an algorithm (using C pseudocode) to delete an element from a stack (where the stack is implemented using a single linked list). Specify its time complexity using Θ notation.
- b) Write non-recursive algorithms (using C pseudocode) for the inorder and preorder traversal of a binary tree having n nodes and specify their best case and worst case time complexities using Θ notation.
- c) Insert the keys in the tree (on the right) arriving in the following sequence: 41, 50, 32 and then delete key 20 from it such that the resulting tree is an AVL tree.



(3+5+2)

- 6 a) Let a sparse matrix be stored in an array $X(0:t, 0:2)$ where t is the number of nonzero elements. The elements $X(0,0)$ and $X(0,1)$ contain the total the number of rows and the total number of columns of the matrix respectively. $X(0,2)$

contains the total number of nonzero elements of the matrix. Let $X(i,0)$, $X(i,1)$ and $X(i,2)$ contain row number, column number and the value of the nonzero element in the matrix respectively, for $1 \leq i \leq t$. Write an algorithm (using C pseudocode) to store two sparse matrices (taken as input from the user) in the form (as specified above using X), perform multiplication of these matrices, store the result in the form (as specified above using X) and display the result (in the form which is usually understood by a user unfamiliar with the representation specified using X). Your algorithm should also be able to notify the user if multiplication is not possible. Also specify the time complexity of your algorithm.

b) Prove that the maximum number of nodes on level i of a binary tree is 2^{i-1} for $i \geq 1$ and the maximum number of nodes in a binary tree of depth k is $2^k - 1$ for $k \geq 1$.

c) For the algorithm *tower* (specified on the right), specify its time and space complexity using Θ notation

Algorithm *tower*(n, source, inter, dest)

```
{
    if(n>0)
    {
        tower(n-1, source, dest, inter);
        print("Move disk", n, "from", source, "to", dest);
        tower(n-1, inter, source, dest);
    }
}
```

(6+2+2)

7 a) Write an algorithm which swaps the left and right children of every node of a binary tree represented using linked representation.

b) Write algorithms (using C pseudocode) to delete an element from a Binary Search Tree (having n nodes) and specify its best case and worst case time complexity using Θ notation.

c) Write a tail recursive algorithm (using C pseudocode) which returns the n^{th} term of the Fibonacci series and specify its time and space complexities using Θ notation.

(4+3+3)

8 a) Write your recursive algorithms (using C pseudocode) to find the Greatest Common Divisor (GCD) and Least Common Multiple (LCM) of two given numbers (provided by the user as input). Specify their time complexities

b) Write algorithms (using C pseudocode) for a menu driven implementation for performing insertion and deletion into a queue (where the queue is implemented using doubly linked list) based on user's choice and find its worst case time complexity using Θ notation.

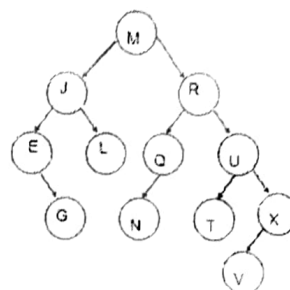
c) For the following, find the function $g(x)$ for which $f(x) = \Theta(g(x))$

i) $f(x) = 2+6x+18x^2/2! + 54x^3/3! \dots + \infty$

ii) $f(x) = 1+x+x^2/2! + x^3/3! + \dots + x^n/n!$

(4+4+2)

9 a) Delete L from this tree (on the right) such that the resulting tree is an AVL tree.



b) Write an algorithm (using C pseudocode) to perform the addition of two polynomials [where each polynomial is stored using a doubly linked list of elements in the form $(m, e_{m-1}, b_{m-1}, e_{m-2}, b_{m-2}, \dots, e_0, b_0)$ where, the first entry m represents the number of non-zero terms followed by two entries (e_i, b_i) for each non-zero term (representing an exponent-coefficient pair) in the order $e_{m-1} > e_{m-2} > \dots > e_0 > 0$] and find the time complexity of your algorithm.

c) Write an algorithm (using C pseudocode) that checks using stacks if a string of parentheses made up of $(,), \{, \}, [, \text{ and }]$ is balanced or not. For example: " $(() f (()) g)$ " is balanced, but " $([)]$ " is not balanced since the parentheses don't match up correctly. The algorithm should return 1 if the input string is balanced and 0 otherwise.

(2+5+3)
